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Relative Speed

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Question 1

A thief steals a car and drives it at 15 km/hr. The theft has been discovered after one hour and the owner of the car sets off in another car at 25 km/hr. When will the owner overtake the thief from the starting point?

A) 1 hour

B) 1.5 hour

C) 2 hour

D) 2.5 hour

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Question 2

P and Q start running in opposite directions (towards each other) on a circular track starting at diametrically opposite points. They first meet after P has run for 75m and then they next meet after Q has run 100 m after their first meeting. Assume that both of them are running a constant speed. The length of the track (in metre) is:

A) 70

B) 175

C) 250

D) 350

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Question 3

X and Y start walking towards each other at 10 am at speeds of 3 km/hr and 4 km/hr respectively. They were initially 17.5 km apart. At what time do they meet?

A) 2:30 pm

B) 11 :30 pm

C) 1:30 pm

D) 12 :30 pm

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Question 4

A hare sees a dog 200m away from her and scuds off in opposite direction at a speed of 24 km/hr. Two minutes later the dog perceives her and gives chase at a speed of 32 km/hr. How soon will the dog overtake the hare, and what distance from the spot from where the hare took flight?

A) 8 min, 2 km

B) 7.5 min, 2 km

C) 7.5 min, 3 km

D) 7.5 min, 2.5 km

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Question 5

Two boys begin together to write out a booklet containing 535 lines. The first boy starts with the first line, writing at the rate of 100 lines an hour; and the second starts with the last line then writes line 534 and so on, backward proceeding at the rate of 50 lines an hour. At what line will they meet?

A) 356

B) 277

C) 357

D) 267

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Question 6

A car of length 4m wants to overtake a trailer truck of length 20m travelling at 36km/hr within 10 seconds. At what speed should the car travel?

A) 12 m/s

B) 14.8 m/s

C) 12.4 m/s

D) 7.6 m/s

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Question 7

The distance between two cities A and B is 80 km. A motorcycle rider starts from A towards B at 7 am at a speed of 10 km/hr. Another motorcyclist starts from B towards A at 8 am at a speed of 25 km/hr. At what time will they cross each other?

A) 10 AM

B) 11 AM

C) 12 PM

D) 9 AM

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Question 8

Two boats travelling at 5 kmph and 10 kmph head towards each other. They begin at a distance of 20 km from each other. How far apart are they (in km) one minute before they collide ?

A) 1/2

B) 1/6

C) 1/4

D) 1/3

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Question 9

Three cars leave A for B in equal time intervals. They reach B simultaneously and then leave for Point C which is 240 km away from B. The first car arrives at C an hour after the second car. The third car, having reached C, immediately turns back and heads towards B. The first and the third car meet a point that is 80 km away from C. What is the difference between the speed of the first and the third car?

A) 60

B) 20

C) 40

D) 80

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Question 10

There are 2 cities P and Q if a car A starts from city P at 6 AM and reaches city Q at noon. another car B starts from city Q at 7 AM and reaches city P at 3 pm another car C starts from city P, an hour earlier than car A if d distance between 2 cities is 240km, and car C can meets car A and car B simultaneously, den how much time would the car C require travelling from city P to city Q?

A) 68/9 hours

B) 68/7 hours

C) 86/9 hours

D) 69/8 hours

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Question 11

The thief Bhagu Ram is spotted by the policeman Pakad Singh from a Distance of 200m. Once they see each other they start running. What is the Distance Bhagu Ram who is running at 5 kmph would have covered before being caught by Pakad Singh running at 7 kmph?

A) 0.5 km

B) 1.5 km

C) 2.5 km

D) 3.5 km

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Question 12

Arun, Barun and Kiranmala start from the same place and travel in the same direction at speeds of 30, 40 and 60 km per hour respectively. Barun starts two hours after Arun.If Barun and Kiranmala overtake Arun at the same instant, how many hours after Arun did Kiranmala start?

A) 3.0

B) 3.5

C) 4.0

D) 4.5

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Question 13

Cities M and N are 600km apart. Bus A starts from city M towards N at 9AM and bus B starts from city N towards M at the same time. Bus A travels the first one-third of the distance at a speed of 40kmph, the second one-third at 50kmph and the third one-third at 60kmph. Bus B travels the first one-third of the total time taken at a speed of 40kmph, the second one-third at 50kmph and the third one-third at 60kmph. When and where will the two buses cross each other?

A) 300 kms from M

B) 280 kms from M

C) 305 kms from M

D) 295 kms from M

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Question 14

Car A trails car B by 50 meters. Car B travels at 45km/hr. Car C travels from the opposite direction at 54km/hr. Car C is at a distance of 220 meters from Car B. If car A decides to overtake Car B before cars B and C cross each other, what is the minimum speed at which car A must travel?

A) 36 km/hr

B) 45 km/hr

C) 67.5 km/hr

D) 18 km/hr

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Question 15

Arun, Barun and Kiranmala start from the same place and travel in the same direction at speeds of 30, 40 and 60 km per hour respectively. Barun starts two hours after Arun.If Barun and Kiranmala overtake Arun at the same instant, how many hours after Arun did Kiranmala start?

A) 3.0

B) 3.5

C) 4.0

D) 4.5

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THANK YOU

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